

LAMPNODE · REQUIREMENTS

LampNode requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	LPN-REQ-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/lampnode		
Source	docs/03-requirements.md @ b3a7651		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

LampNode requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with a lighting authority or users, and will be checked by calculation at TRL 3 and revised after co-design (see LPN-PRB-001). The status column gives the TRL 2 position from the estimates in LPN-PRC-001.

The **reference case** used throughout is a 100 W LED cobra-head luminaire with a 0 to 10 V driver, mounted about 7.8 m above a residential street, lit about 4,100 h per year (about 11.2 h per night on average).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 2 status
R1	Fit the existing socket	Plugs into an ANSI C136.41 7-contact twist-lock receptacle; replaces a photocontrol with no tools and no luminaire rewiring in 5 min or less at height	Dimensional check against the standard; fit trial on a donor luminaire	Met by design (proposed socket)
R2	Mains supply range	120 to 277 V AC nominal, 50 or 60 Hz, operating from 100 to 305 V	Power supply datasheet; design review	Met for 120 to 277 V; not met for 347 V and 480 V circuits
R3	Switch the luminaire	Switch LED luminaire loads up to 400 W at 277 V, with a relay rated for the driver's inrush current, for 100,000 or more cycles	Relay datasheet against driver inrush data	Met by selection, inrush to be confirmed per driver
R4	Dim the luminaire	0 to 10 V dimming output, 10 to 100 % in 1 % steps, sinking at least 10 mA, isolated from mains	Circuit review; bench check at TRL 4	Met by design; DALI-2 D4i variant not in base design
R5	Schedule without the network	Astronomical clock (dusk and dawn from latitude and longitude) and at least 8 dimming steps per night; keeps running with no network for 30 days or more, clock drift 2 min or less	Firmware sketch review; RTC drift calculation	Met by design
R6	Detect presence	Detect a walking pedestrian at 15 m or more and a cyclist or car at 25 m or more along the street; raise the lamp to full within 1 s; hold for a set time (default 60 s)	Radar range calculation; field trial	At risk , radar range from 7.7 m height unverified
R7	Save energy	35 % or more annual energy reduction versus photocell full-power operation in the reference case, controller use included	Energy calculation (LPN-PRC-001)	Met by estimate: about 41 %
R8	Measure energy	Active power within ± 2 % from 10 to 400 W; also voltage, current, power factor and cumulative energy; reported every 15 min	Metering IC datasheet; calibration method at TRL 3	Met by selection, unverified
R9	Report faults	Detect lamp out, day burning, cycling, tilt or knock-down, and loss of mains; report within 30 min; the server flags a node silent for 1 h	Fault logic review	Met by design
R10	Low self-consumption	1.0 W or less average from the mains, sensor head included, expansion ports unloaded	Power budget	Met by estimate: about 0.6 W, 0.8 W with margin
R11	Survive the environment	IP66; operate from -30 to +70 °C; UV-stable housing; proposed surge target 10 kV / 5 kA combination wave, final level taken from ANSI C136.2 at TRL 3	Datasheets and design review	Unverified
R12	Fail safe	Lamp on at night if the controller, firmware or network fails; lamp on within 2 s of power-up at dusk without the network; open 0 to 10 V line gives full output	Circuit and firmware review	Met by design; driver open-line behavior to be confirmed

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 2 status
R13	Host other sensors	Two sealed M12 expansion ports with the FieldNode pinout (proposed), 12 V SELV, 3 W total, cable up to 10 m	Power budget; design review	Met at night; not met in daytime on cabinet-switched feeders
R14	Privacy	Presence only: no images, audio or personal identifiers are captured or leave the device; optional 15 min presence counts	Design review; open firmware	Met by design
R15	Secure and open	LoRaWAN 1.0.4 or later with AES-128 session keys; signed firmware; published payload format; works with any LoRaWAN network server	Firmware sketch review	Met by design; TALQ bridge not met at TRL 2
R16	Affordable	Controller and one sensor head \$150 or less in parts at prototype quantities	Priced BOM (bom/ bom.csv)	Met: about \$124

■ Requirements not met or at risk

- **R2:** 347 V (common in Canada) and 480 V lighting circuits need a different power supply and relay; not covered.
- **R4:** only 0 to 10 V dimming in the base design; D4i luminaires would need a DALI-2 variant.
- **R6:** presence range is the least certain figure in the concept.
- **R13:** on feeders switched at a cabinet, hosted sensors lose power in daytime unless they carry their own storage.
- **R15:** no TALQ bridge yet; interoperability rests on open LoRaWAN and a published payload.
- **R8 and R11** are unverified until TRL 3 and later testing.

■ Assumptions

- Dusk-to-dawn operation of about 4,100 h per year, a mid-latitude average; high latitudes differ strongly by season.
- Luminaire power is taken as proportional to commanded light output; real drivers are less efficient when dimmed, so savings may be a few percentage points lower.
- Reference dimming profile (proposed, for estimates only): full output from dusk to 22:00, then a 30 % floor with presence boosts to full from 22:00 to 06:00, and full output from 06:00 to dawn when dawn is later. Presence boosts are assumed to occupy 15 % of the dimmed hours on a residential street (estimate).
- For comparison, a schedule-only profile (full output except 50 % from 23:00 to 05:00, no presence sensing) is also estimated in LPN-PRC-001.
- The road owner sets the floor level and hold time. Nothing here proposes a light level for any road.

Safety: LampNode connects to mains voltage and is installed at height beside traffic. Any requirement change that affects fail-safe behavior (R12) or light levels must be reviewed with the road owner.